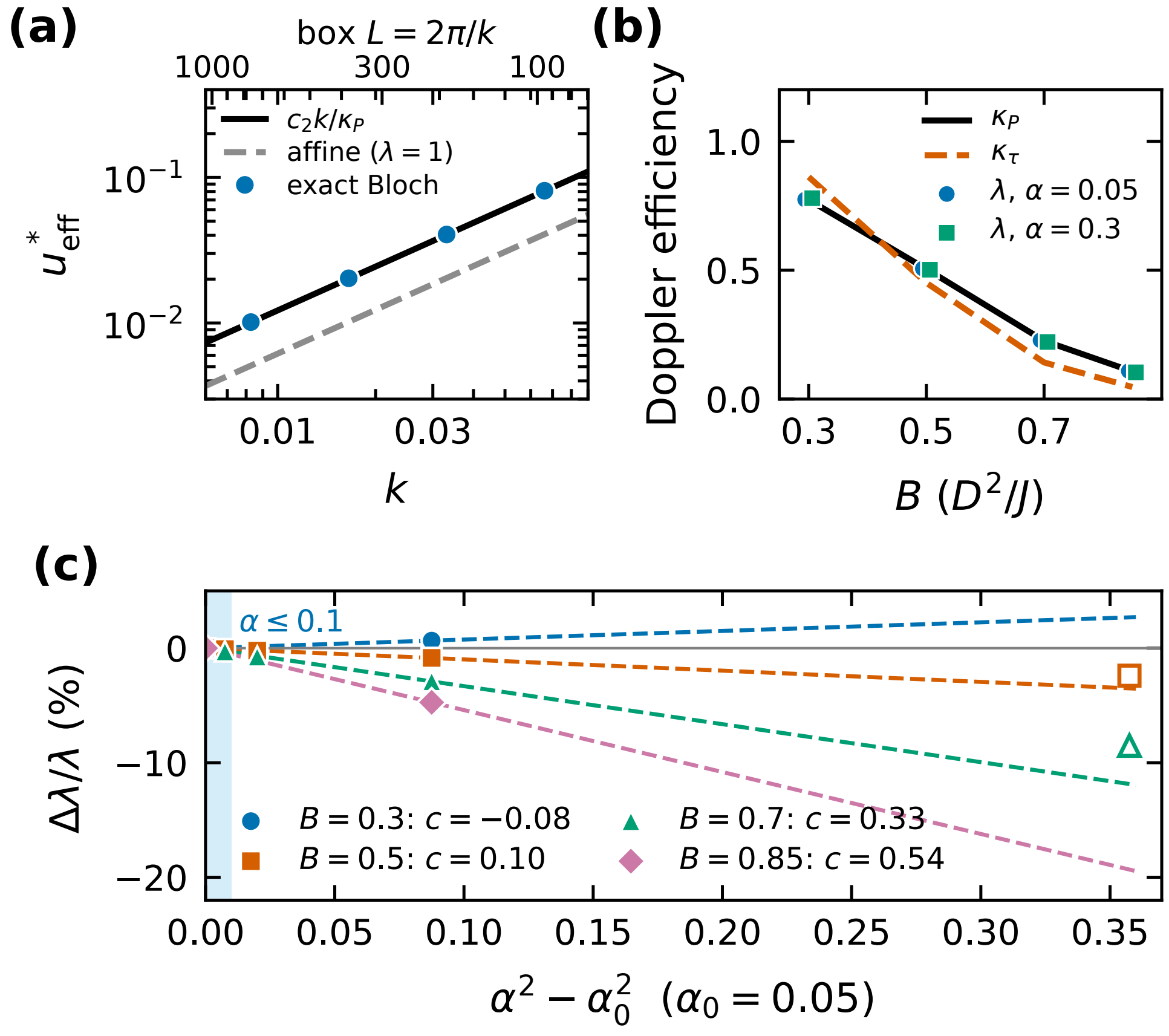
(a)
box $L = 2\pi/k$
1000
300
100
$c_2 k/\kappa_P$
affine ($\lambda = 1$)
exact Bloch
10^{-1}
10^{-2}
u^*_{eff}
0.01
0.03
k
(b)
κ_P
κ_τ
λ, $\alpha = 0.05$
λ, $\alpha = 0.3$
1.0
0.5
0.0
Doppler efficiency
0.3
0.5
0.7
B (D^2/J)
(c)
$\alpha \le 0.1$
0
−10
−20
$\Delta\lambda/\lambda$ (%)
$B = 0.3$: $c = -0.08$
$B = 0.5$: $c = 0.10$
$B = 0.7$: $c = 0.33$
$B = 0.85$: $c = 0.54$
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
$\alpha^2 - \alpha_0^2$ ($\alpha_0 = 0.05$)